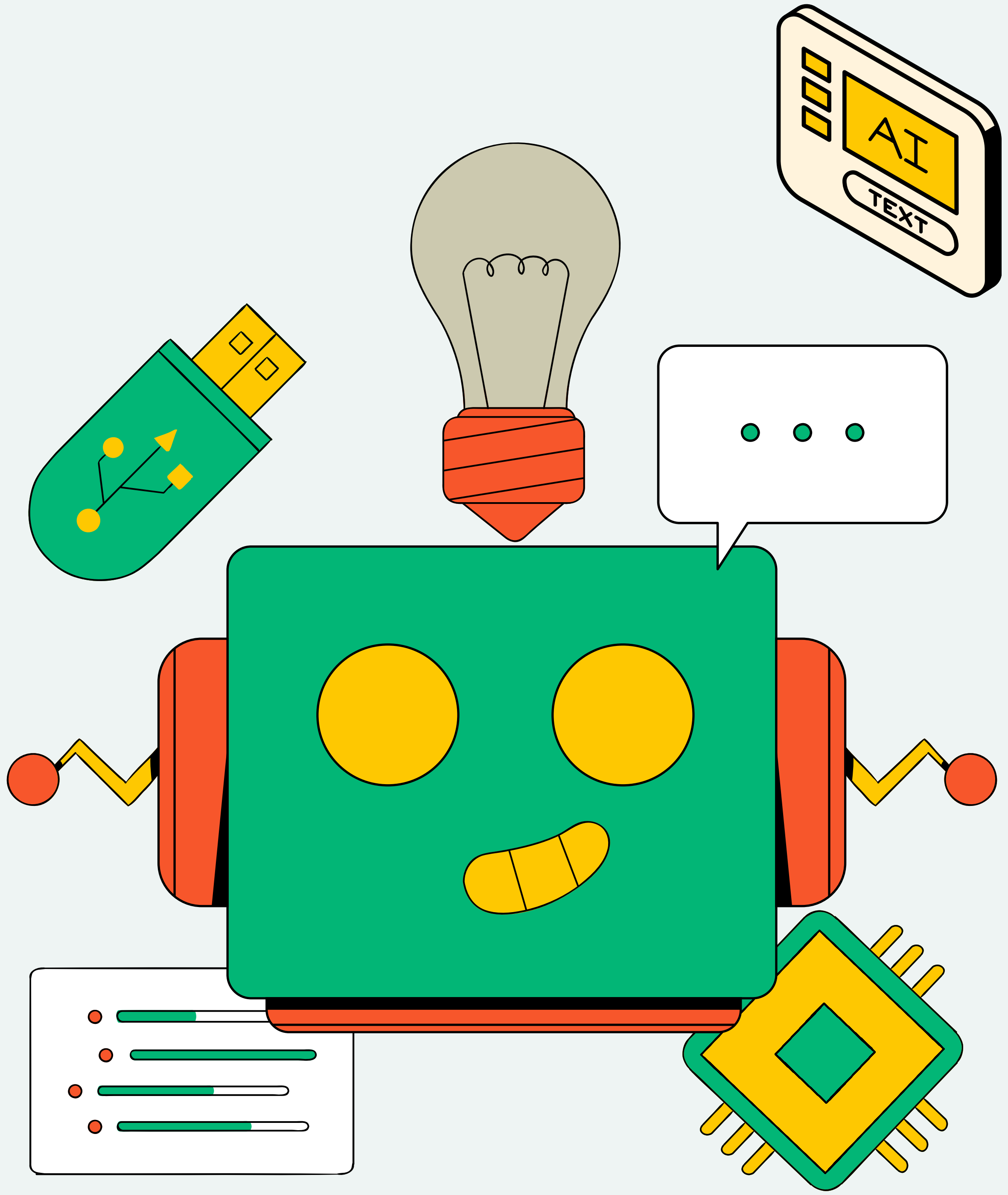
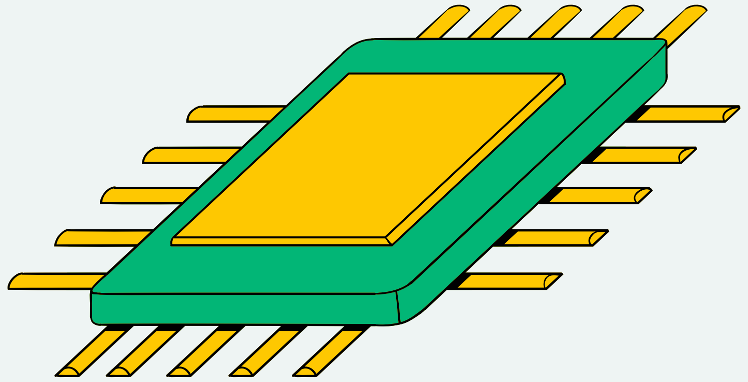


AI SURVEILLANCE AND PRIVACY: NAVIGATING THE BALANCE

PRESENTATION

PRESENTED BY:

GROUP 8



PRESENTATION OUTLINE

- Introduction
- Research Findings
- Balance + Evaluate Public Safety and Privacy
- AI in Big Data and Surveillance
- Implementation Plan

  Phase 1: Initial Deployment

  Phase 2: Enhancements and Extentions

  Phase 3: Consolidation and Sustainabiity

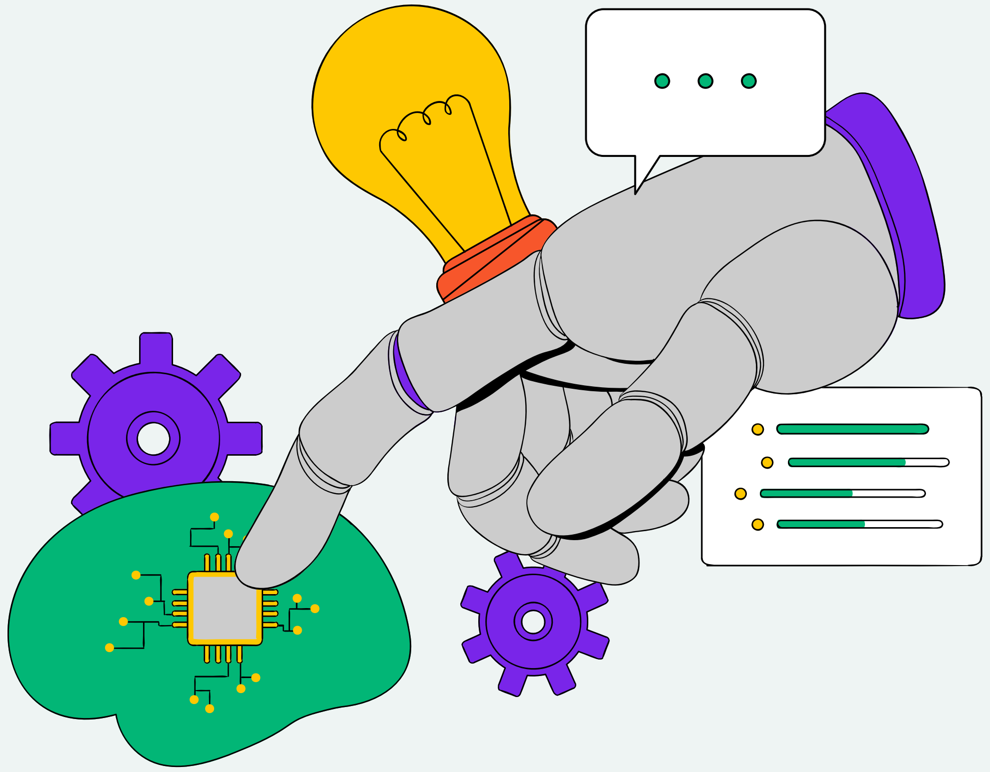
- Objectives and Measurement
- Methodology and Reflection
- Preparing for the Future

INTRODUCTION

Introducing the swift progress in AI that powers surveillance systems, focusing on how these innovations are revolutionizing urban security.

- **Enhancing Urban Security:** AI systems bolster emergency response, crime prevention, and traffic management, promising a safer urban environment.
- **Privacy vs. Security:** The expansion of surveillance raises significant privacy concerns, highlighting the need for a balanced approach that respects individual rights.
- **The Big Data Challenge:** Managing and protecting vast amounts of personal data without infringing on privacy presents a complex dilemma.
- **Economic and Social Considerations:** High costs, public skepticism, and potential regulatory hurdles underscore the need for careful implementation and public engagement.
- **Research Objective:** To explore the effective use of AI surveillance in improving urban safety while ensuring privacy protection and overcoming associated challenges.

RESEARCH FINDINGS



- AI increases public area surveillance efficiency and capabilities.
- Monitoring real-life footage in real time boosts the efficiency of public officials (Montasari, 2024).
- Facial recognition technology determines identity, thus enhancing the accuracy of surveillance.
- UAVs act as security cameras by substituting human surveillance.
- AI algorithms can detect anomalies and analyze crowd behavior (Thakur et al., 2021).
- Predictive policing makes use of AI–driven data analytics.
- AI surveillance systems are confronted with problems of law and morality (Naik et al., 2022).

STRATEGIES FOR BALANCING PUBLIC SAFETY AND PRIVACY

- **Regulatory Procedures & Privacy Rules:** Implementing standards for data collection, storage, and use. Requires consent for surveillance coverage.
- **AI Privacy Innovations:** Enhancing data security through encryption, data anonymization, and privacy-preserving technologies to mitigate unauthorized access risks.
- **Transparency & Accountability:** Establishing independent auditing, public enforcement mechanisms, and reporting to build trust and demonstrate commitment to responsible surveillance.
- **Stakeholder Collaboration:** Fostering partnerships between governments, industry, civil society, and tech experts for shared solutions that respect both security and privacy.
- **Public Education & Awareness:** Encouraging informed choices on AI surveillance through educational campaigns, fostering a society that is both informed and vigilant.

EVALUATING THE OPTIONS FOR PRIVACY-SECURITY BALANCE

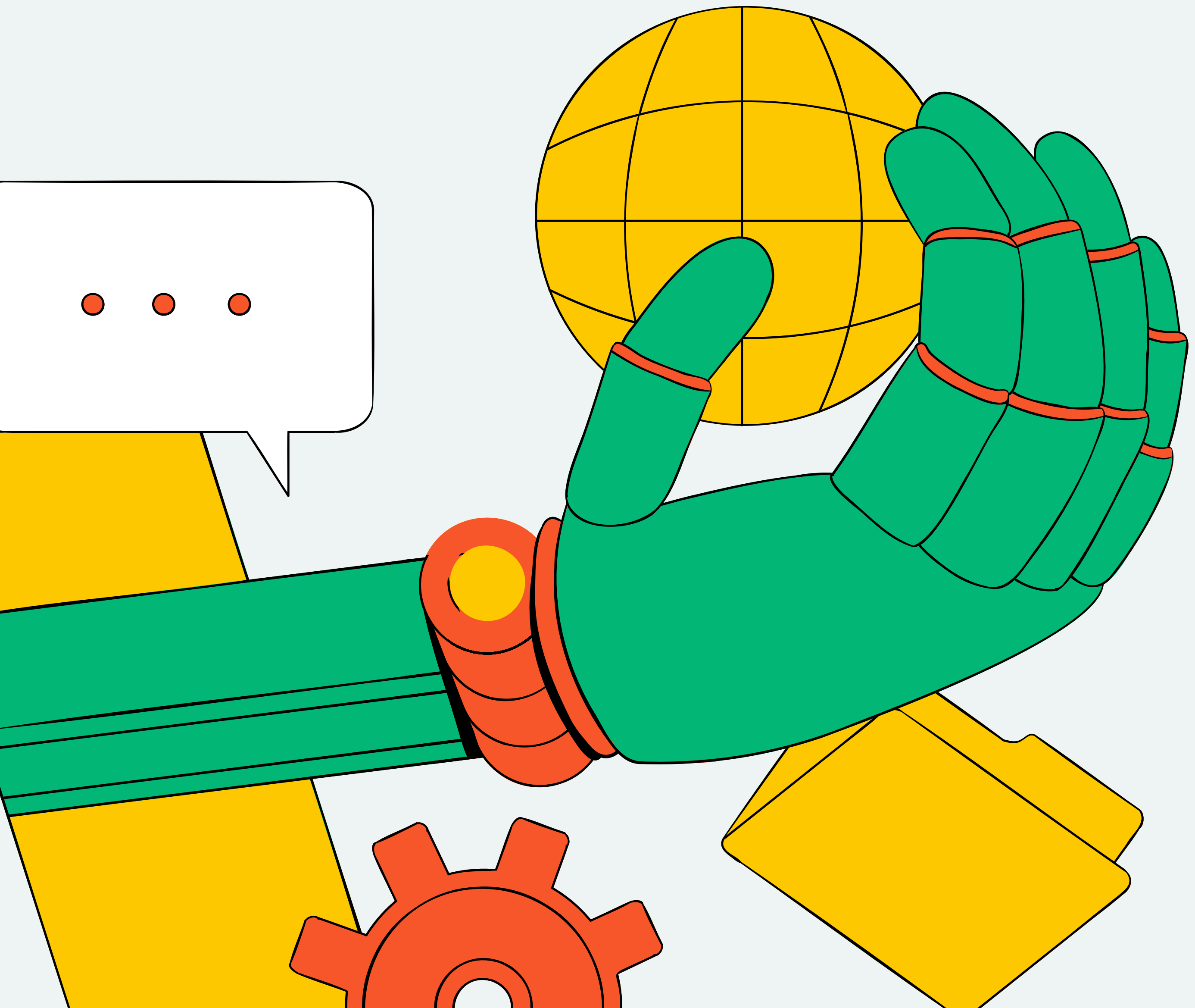
- **Regulatory & Privacy Standards:**
  - **Benefits:** Enhances public trust and transparency.
  - **Downsides:** Potential hindrance to innovation; ineffective in low enforcement areas.

- **Privacy-Enhancing Technologies:**
  - **Benefits:** Improves privacy without compromising security.
  - **Downsides**: Complex to implement; expensive upgrades required.

- **Accountability & Transparency:**
  - Benefits: Promotes public discourse and oversight.
  - Downsides: Requires substantial resources; effectiveness limited without enforcement.

- **Collaborative Approaches:**
  - **Benefits:** Leverages collective expertise for balanced solutions.
  - **Downsides:** Challenging to achieve consensus; requires ongoing commitment.

- **Public Education Efforts:**
  - **Benefits:** Builds a vigilant, engaged society.
  - **Downsides:** Limited by accessibility and the effectiveness of messaging; costly long-term effort.



DECISION MAKING

Artificial Intelligence in Surveillance

Artificial Intelligence and Big Data Analytics

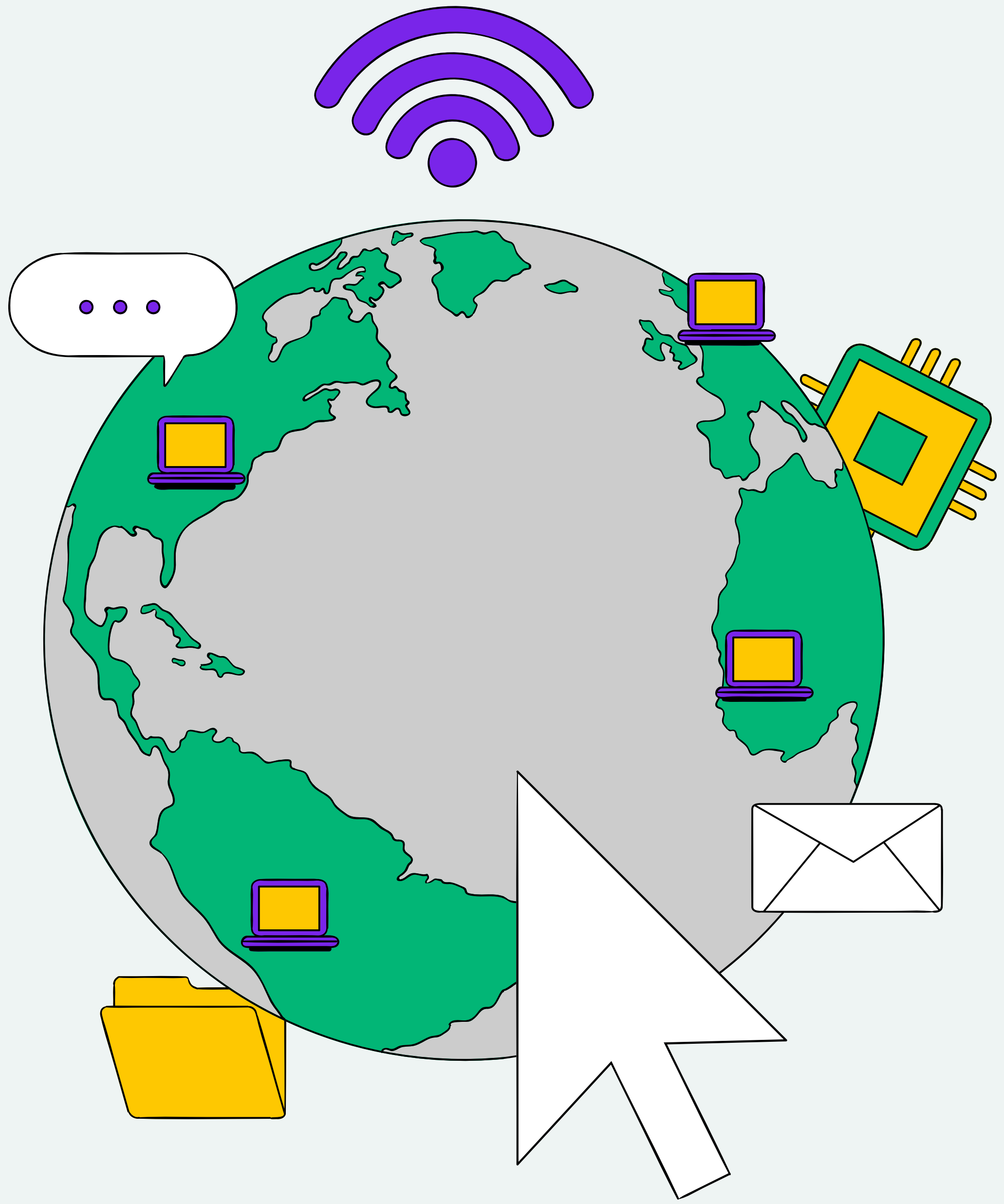
Extensibility

Data security

Flexibility

Privacy security

ARTIFICIAL INTELLIGENCE IN SURVEILLANCE



- Artificial Intelligence (AI) can drastically improve social security and people's safety regarding surveillance in public places.
- AI technology can automate identifying and finding suspicious behaviours or security risks through behavioural and facial analyses.
- AI Surveillance technology enhances the real-time nature of surveillance and improves and prevents the ability to handle emergencies.

ARTIFICIAL INTELLIGENCE AND BIG DATA ANALYTICS

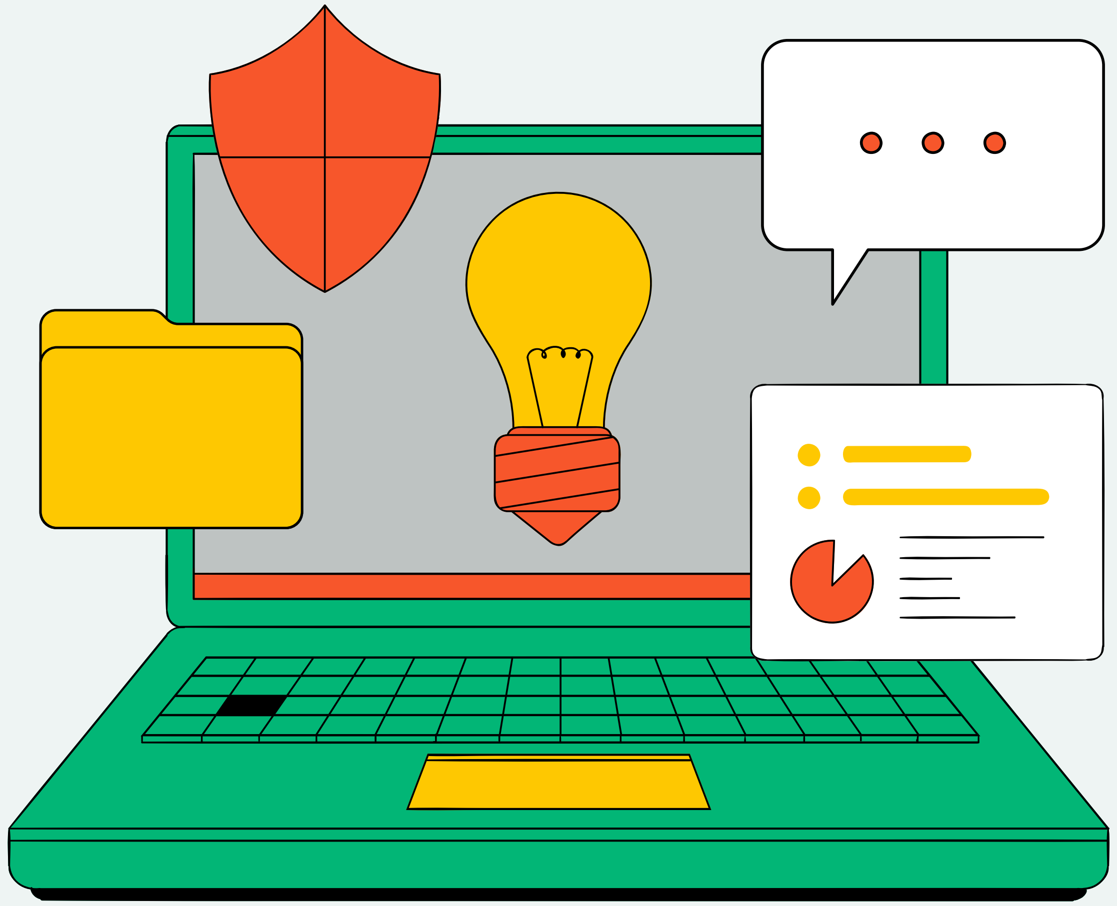
- AI’s role in surveillance is not just reactive but also proactive. By analyzing and learning from vast amounts of historical data, AI can enable predictive policing.

- Based on behavioural patterns and historical data, these analytics can effectively alert law enforcement officers to potential security threats, giving them ample time to take preventive measures.

- This proactive approach to surveillance enhances security and instills a sense of safety in the public.

EXTENSIBILITY AND FLEXIBILITY

- The use of AI in surveillance does not stop at surveillance cameras. AI can adapt to varying sizes and complexity of surveillance needs.
- AI technology combined with unmanned aerial vehicles (UAVs) can dramatically reduce the dead space of surveillance in the city. UAVs can monitor hard-to-reach places or vast areas and analyze the images through AI.
- This type of surveillance can be used flexibly in different scenarios, thus easing police forces and providing a safer way for police officers to monitor.



DATA SECURITY AND PRIVACY SECURITY

- Equipping AI with sophisticated encryption algorithms and anonymization features can ensure that sensitive information is processed while protecting personal privacy.

- Increased transparency and accountability can also help increase people's trust. Work with stakeholders to develop surveillance programs safeguarding privacy and gaining people's trust.

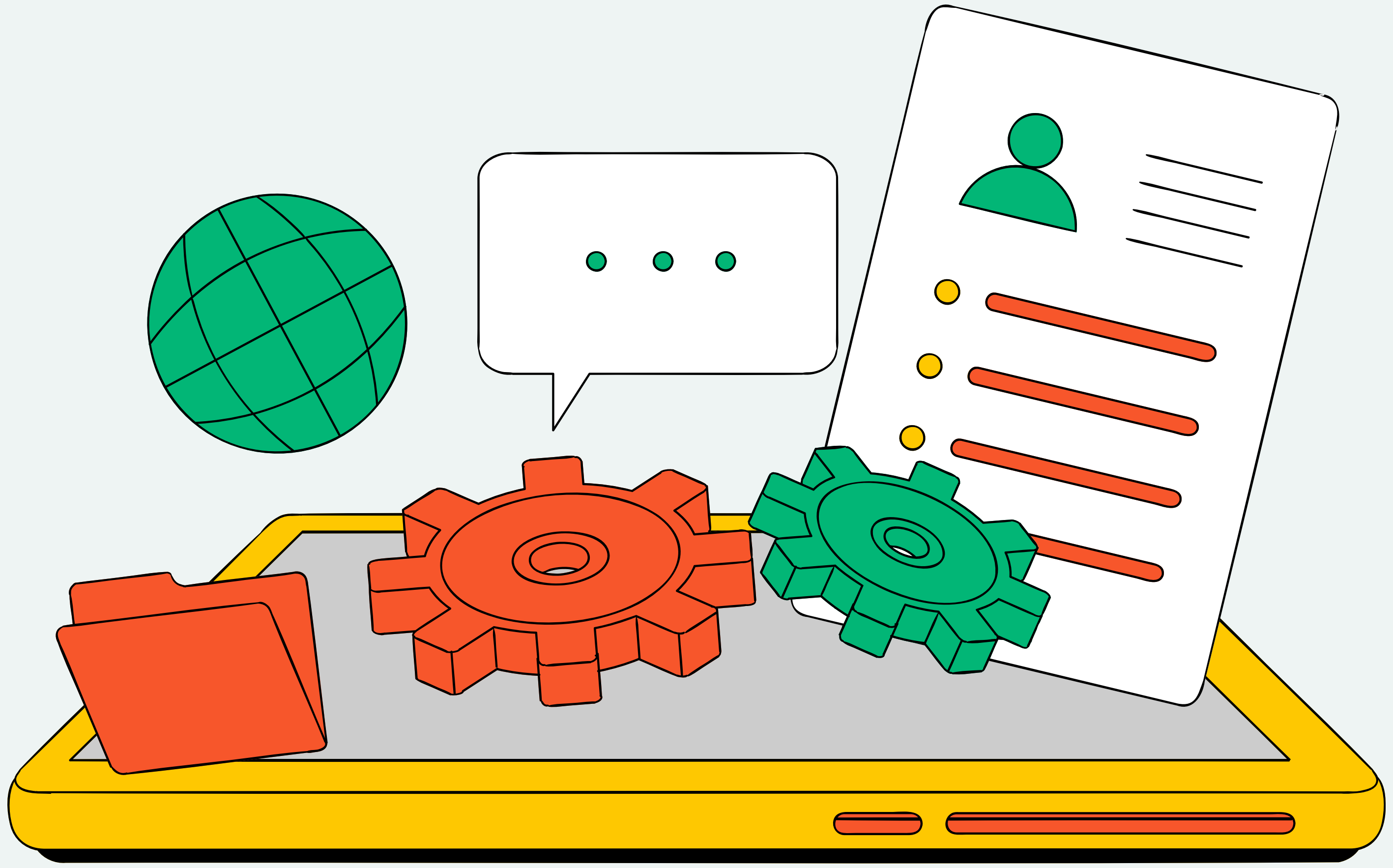
- Increase social awareness of AI and provide more relevant education to the public. This can also help build public trust in AI surveillance systems and promote participation.

PART6-IMPLEMENTATION PLAN

Phase1-Initial deployment

Phase2-Enhancements and Extensions

Phase3-Consolidation and Sustainability



PHASE1-INITIAL DEPLOYMENT

OBJECTIVE:
ESTABLISH INFRASTRUCTURE FOR AI MONITORING SYSTEM.

KEY ACTIONS:

1. GATHER PUBLIC INPUT AND DEPLOY IN HIGH-CRIME OR SAFETY-ISSUE AREAS.
2. INSTALL AI CAMERAS WITH FACIAL RECOGNITION AND BEHAVIORAL ANALYSIS.
3. EXPAND SURVEILLANCE RANGE WITH MOBILE INTEGRATED UAVS.

PRIVACY MEASURES:

1. IMPLEMENT PRIVACY PROTECTION FOR RESIDENTS IN PATROL AREAS.
2. USE ADVANCED ENCRYPTION ALGORITHMS AND DATA ANONYMIZATION PROTOCOLS.

PHASE2- ENHANCEMENTS AND EXTENSIONS

OBJECTIVE: IMPROVE THE SURVEILLANCE SYSTEM'S CAPABILITIES AND EXPAND COVERAGE.

KEY ACTIONS:

1. ENHANCE THE ACCURACY OF FACIAL AND BEHAVIORAL RECOGNITION.
2. FORMULATE DATA COLLECTION, STORAGE, AND USE GUIDELINES.
3. CONDUCT PUBLIC EDUCATION ACTIVITIES AND PROVIDE FEEDBACK MECHANISMS.

PRIVACY MEASURES:

1. ESTABLISH PARTNERSHIPS BETWEEN GOVERNMENT AGENCIES, AI PROVIDERS, AND CIVIL SOCIETY ORGANIZATIONS.

PHASE3- CONSOLIDATION AND SUSTAINABILITY

OBJECTIVE: ENSURE LONG-TERM SUSTAINABILITY AND TRANSPARENCY OF THE AI MONITORING SYSTEM.

KEY ACTIONS:

1. ESTABLISH REGULAR REPORTING MECHANISMS AND AN INDEPENDENT OVERSIGHT BODY.
2. CONTINUOUSLY UPDATE THE AI MONITORING SYSTEM WITH THE LATEST TECHNOLOGY.
3. CONDUCT PERIODIC EVALUATIONS TO ASSESS IMPACT ON PUBLIC SAFETY AND PRIVACY.

COMMUNITY INVOLVEMENT:

1. ENHANCE DATA TRANSPARENCY AND COMMUNITY PARTICIPATION.

PART 7: OBJECTIVES AND MEASUREMENT

- Public Safety
- Personal Right (Privacy)
- Crime Rate
- Costs
- AI Algorithm
- Legal Compliance

OUR OBJECTIVES

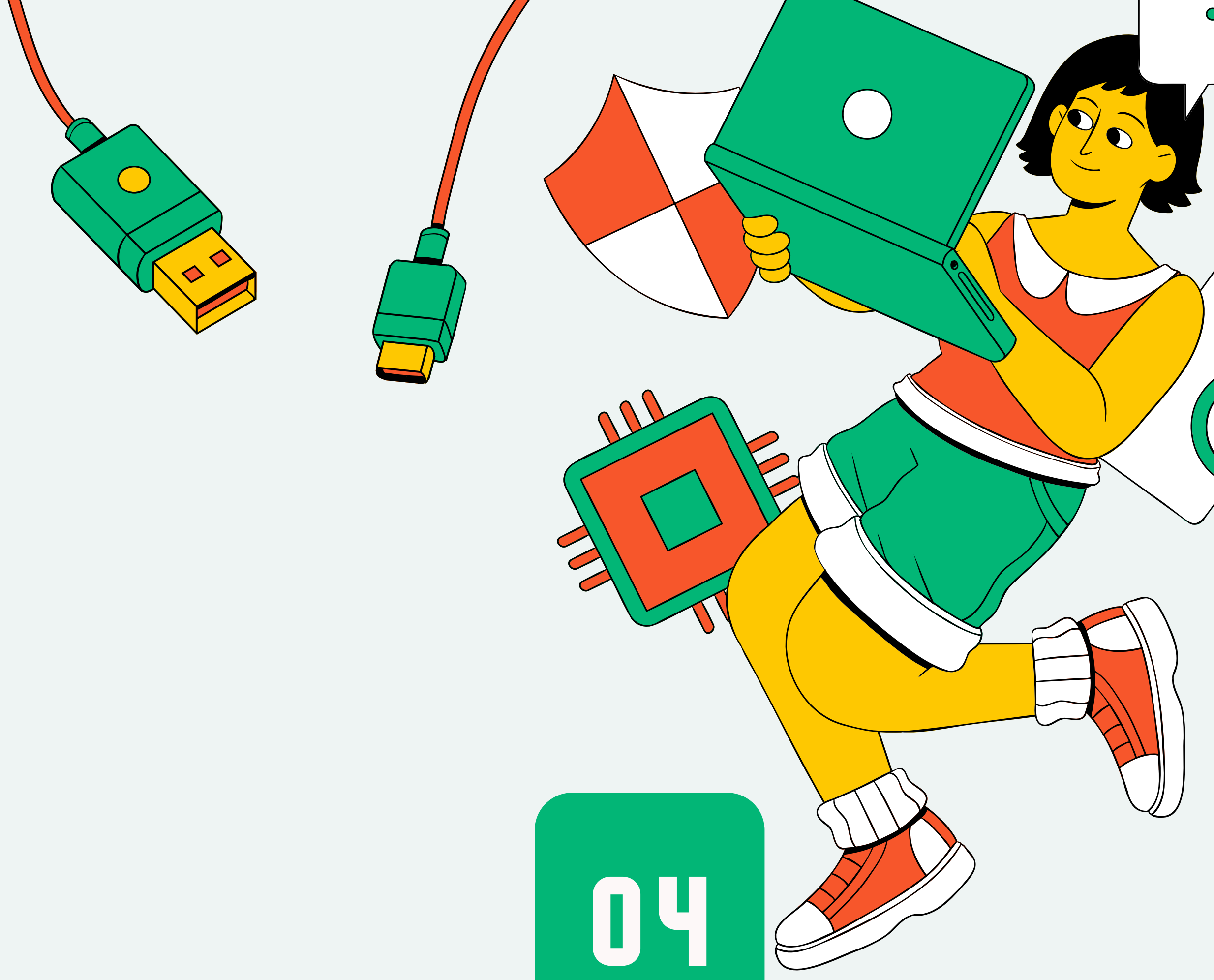
IMPROVE SURVEILLANCE

- Apply AI Technology
- Proactively identify suspicious behaviors and safety risks
- Overcome low efficiency, passive issues

BALANCE PRIVACY RIGHTS

- Stricter accountable rules
- More transparency
- Government involvement
- Public education

MEASUREMENTS



01

CRIME RATE

Comparing the crime rate changes in specific periods and locations (Scope and Time)

The decrease in crime rate shows positive improvement in public safety

02

COSTS

Comparing the costs per incident in the long term (AI implementation has a high one-time investment which makes high initial costs)

The lower costs show better cost perfromance

03

AI ALGORITHM

The AI Algorithm Accuracy Based on the true and false detection rate, detection time, and ethical considerations

High truth, detection rate, and low detection time show positive performance.

04

LEGAL COMPLIANCE

The number of legal compliance cases including public surveys, legal cases, cyber-attacks, and actual data disclosed times

The lower chance of these incidents happen show better implementation

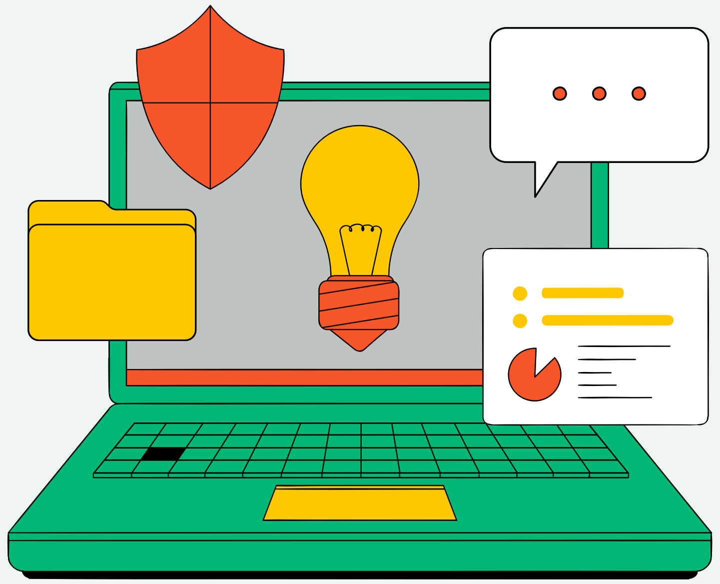
METHODOLOGY AND REFLECTIONS

Process Overview:

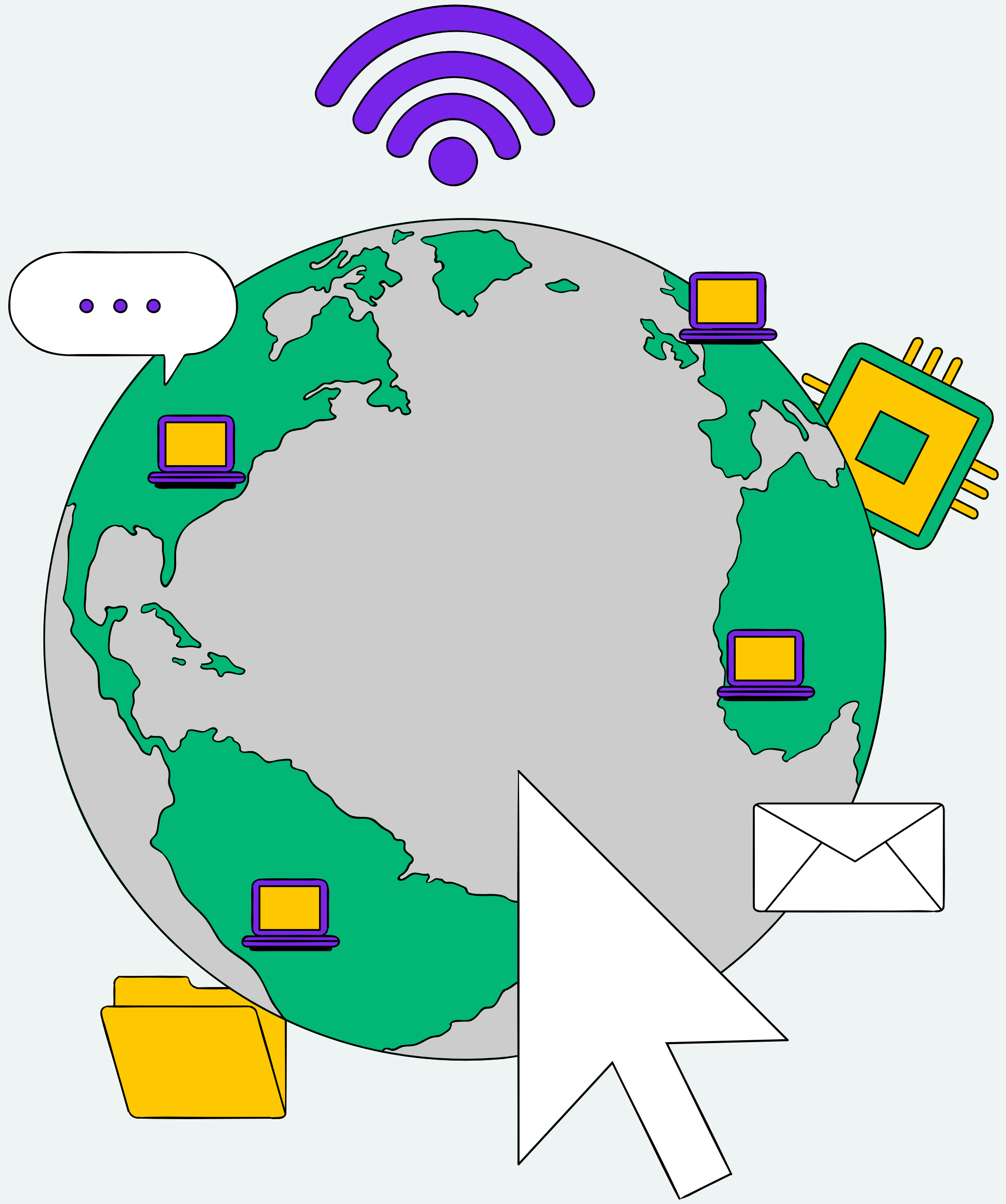
- Deliberate topic selection process using Slack and Zoom meetings.
- Crafting the project involved collaborative analysis and outlining of article structure using Google Doc.
- Strategic organization ensured coherence and efficiency throughout the essay.
- Communication channels included Zoom meetings for discussions and Slack for logistical matters.

Reflections:

- Collective growth and learning through diverse viewpoints and approaches.
- Challenges included scheduling mutual availability for Zoom meetings, emphasizing effective communication and empathy.



PREPARING FOR THE FUTURE



Preparing for the future with artificial intelligence (AI) involves strategic thinking, adaptability, and a focus on harnessing the potential of AI to drive innovation and address societal challenges.

Here are some key steps and considerations for preparing for the future with AI:

- Education and Skills Development
- Data Management and Security
- Ethical and Responsible AI
- AI Integration and Talent
- Innovation and Culture

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